GPR learning exercise

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Why I did this

To prepare my application for the EngD on standardizing GPR fieldwork and data management, I wanted to move from a theoretical understanding of GPR to a first practical experience with real data.

What I did

- Processed a real GPR profile (the GSSI example file included in the open-source software GPRPy; a 20.8 m line with a 48 ns time window) in four steps: dewow, background removal, gain, and conversion from time to depth with an assumed wave speed.
- Made a simple simulation of two pipes, 0.6 m and 1.0 m deep, in dry sand and in wet clay.
- Read the file header and built a metadata form in Excel with 17 fields in five groups: location and time, equipment, ground conditions, processing, and verification.

What I observed

- Processing removed the strong horizontal band near the top (most likely the direct wave between the antennas) and made a sloping layer and deeper reflections much clearer. Processing does not make the radar reach deeper; it makes weak signals from depth visible.
- The processing parameters (the dewow window, the number of traces used for background removal, the gain power and the wave speed) are choices made by the person who processes the data. Changing them changes the image, and a wrong wave speed gives a wrong depth. GPRPy saved all steps and parameters automatically in a history file.
- In my simplified simulation, assumed differences in wave speed and attenuation between dry sand and wet clay made the same pipes appear later and much weaker, and the deeper pipe almost disappeared. The simulation is not a validated physical model, but it illustrates why soil type and moisture need to be recorded.
- The file header recorded the device settings (time window, trace spacing, samples per trace), but none of the ground conditions. In my form, 10 of the 17 fields were missing and one was only partly known (the make of the device could be inferred from the file format, but not the model). All four ground-condition fields were missing. These are the fields that an operator has to record on site.

What this means for a fieldwork protocol

Without a record of the ground conditions and the processing choices, it is not possible to know whether a difference between two surveys comes from the device, from the soil or from the processing. A protocol should make these fields a fixed part of every survey, so that measurements from different systems and conditions can be compared.

What I still need to learn

Converting travel time into depth is simple (depth = speed × time / 2), but the result is only as good as the wave speed. I would like to learn how the speed is best estimated on site, for example from the shape of the hyperbolas or by calibration at a point of known depth, how it changes when the soil changes along one line, and how its uncertainty could be recorded in the metadata so that a depth is never reported without it. I also want to gain experience with real GPR equipment, other detection methods, and Dutch standards such as KLIC and IMKL.